



MEDIACREST
TRAINING COLLEGE

Advanced Data Analytics & Machine Learning

Outlier Detection, Statistical Thinking & Data Normalization –
Week 2, Day 1

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Abstract

This session transitions from healthcare data to the financial sector, applying rigorous preprocessing to a **Kenyan SACCO/Bank Churn Dataset** of 10,000 customers.

- **Outlier detection:** contrasting visual boxplots with statistical methods (IQR and Z-score) to identify extreme financial balances.
- **Treatment philosophy:** distinguishing between data-entry errors and valid high-net-worth extremes, applying **capping (Winsorization)** to preserve sample size.
- **Feature encoding:** matching strategies to data cardinality — Ordinal for hierarchies, One-Hot for nominal, and **K-Fold Target Encoding** for high-cardinality variables to prevent leakage.
- **Scaling & normalization:** evaluating Standard, MinMax, and Robust scalers, and clarifying algorithm-specific sensitivity and the “normality myth.”
- **Statistical thinking:** building a transferable preprocessing mindset that prioritizes domain context over mechanical rule-following.

Philosophy: *Understand the data, identify the problem, select the transformation, justify the decision.*

Session Objectives

- **Detect** outliers visually and statistically using IQR and Z-score methods on skewed financial variables.
- **Differentiate** between impossible errors and valid extreme values using domain knowledge.
- **Apply** capping (Winsorization) to limit the influence of extremes without destroying sample size.
- **Select** appropriate encoding strategies (Ordinal, One-Hot, Target) based on variable cardinality and hierarchy.
- **Prevent** target leakage in high-cardinality encoding using K-Fold cross-validation techniques.
- **Evaluate** scaling algorithms (Standard, MinMax, Robust) and align them with the mathematical requirements of specific ML models.

Agenda

- 1 **Part 1: Data Loading & The Preprocessing Mindset** (Dataset overview, universal workflow, key characteristics)
- 2 **Part 2: Outlier Detection** (Visual boxplots, IQR method, Z-score method)
- 3 **Part 3: Outlier Treatment Decisions** (Error vs. extreme, decision framework, capping mechanics)
- 4 **Part 4: Feature Encoding Strategies** (Ordinal, One-Hot, Target encoding, leakage prevention)
- 5 **Part 5: Scaling & Statistical Thinking** (Algorithm sensitivity, scaler comparison, the normality myth)

Dataset

Kenyan SACCO/Bank Churn Dataset: 10,000 customers, 20 variables including demographics, financial balances, engagement metrics, and the target variable (Churn).

Part 1: Data Loading & The Preprocessing Mindset

"The dataset may change, but the preprocessing logic does not."

Kenyan SACCO/Bank Churn Dataset

Key characteristics

- **10,000 customer records** from a retail financial institution.
- **20 variables:** demographics, account details, engagement.
- **Target variable:** Churn (1 = exited, 0 = retained).
- Mix of numerical, categorical, and binary features.

Key variables for analysis

- **Financial:** Balance, UtilizationRate, SaccoScore
- **Demographics:** Age, Gender, Education, MaritalStatus
- **Engagement:** Tenure, MonthsInactive, BranchVisits
- **Products:** NumProducts, HasCrCard, HasLoan

Why This Dataset?

Financial data presents unique challenges: extreme right-skewness in balances, high-cardinality categoricals (e.g., 47 counties), and complex customer behaviors. It perfectly tests advanced preprocessing techniques.

Prediction Task

Classify whether a customer will churn based on their financial health, demographic profile, and engagement history.

The Universal Preprocessing Mindset

Transferable principles

Whether analyzing MOH clinical records or SACCO financial data, the core statistical concepts remain identical. The goal is not just to “clean” data, but to develop a systematic mindset.

The Universal Workflow

- 1 **Inspect:** Profile distributions and data types.
- 2 **Identify:** Flag missingness, outliers, and skewness.
- 3 **Clean:** Handle errors and impossible values.
- 4 **Transform:** Address skewness and scale features.
- 5 **Encode:** Convert categoricals to numerical formats.
- 6 **Validate:** Ensure no leakage or information loss.
- 7 **Model:** Feed model-ready data to algorithms.

Domain Agnosticism

A heavily skewed financial balance requires the same mathematical treatment (e.g., log transform or robust scaling) as a heavily skewed clinical lab result. The *context* changes, but the *math* does not.

The Ultimate Goal

Produce reliable, representative, and model-ready data without unnecessarily destroying useful business information.

Key Data Characteristics to Inspect

Before applying any transformations, we must categorize the variables by their statistical behavior.

Characteristic	Example Variables	Preprocessing Implication
Skewed Numerical	Balance, Income	Require log transformations or robust scaling to handle long right tails.
High-Cardinality	County, BranchID	One-hot encoding creates too many columns; requires Target Encoding.
Ordinal Categorical	Education Level	Has a natural hierarchy; requires Ordinal Encoding.
Nominal Categorical	Gender, AccountType	No natural order; requires One-Hot Encoding.
Impossible Values	Age = 150, Tenure < 0	Must be investigated and corrected or removed as errors.

Data Types & Consistency

Ensure categorical, numerical, and identifier variables are correctly typed. Incorrect types (e.g., numeric IDs stored as integers instead of strings) cause silent errors during grouping and encoding.

Part 2: Outlier Detection

"Is it an error, or is it just unusual?"

Visual Outlier Detection: Boxplots

The Boxplot Anatomy

- **Box:** Represents the Interquartile Range (IQR) — the middle 50% of the data (Q1 to Q3).
- **Line inside box:** The median (Q2).
- **Whiskers:** Extend to the furthest points within $1.5 \times \text{IQR}$ from the box.
- **Dots outside whiskers:** Potential outliers.

Observations in our Data

- **Balance:** Shows a massive right tail with hundreds of dots far beyond the upper whisker. Highly skewed.
- **Age:** Relatively normal distribution with very few, if any, extreme outliers.

Why Visualize First?

Summary statistics (mean, std) hide the shape of the distribution. A boxplot immediately reveals skewness and the sheer volume of extreme values before you commit to a mathematical formula.

Financial Context

In banking, a balance of 5,000,000 KES might look like a statistical outlier, but visually, it's part of a continuous long tail of high-net-worth clients, not an isolated data-entry glitch.

Statistical Detection: The IQR Method

The Formula

- $Q1 = 25\text{th percentile}$
- $Q3 = 75\text{th percentile}$
- $IQR = Q3 - Q1$
- **Lower Bound:** $Q1 - 1.5 \times IQR$
- **Upper Bound:** $Q3 + 1.5 \times IQR$

Results for Account Balance

- Lower Bound: -199,286 KES
- Upper Bound: 450,070 KES
- **Outliers flagged: 503**

Why IQR?

The IQR method is **robust to skewness and outliers**. Because it relies on quartiles (medians of halves) rather than the mean and standard deviation, extreme values do not pull the boundaries outward.

Interpretation

Any customer with a balance above 450,070 KES is flagged. In our dataset, 503 customers exceed this threshold. Are they errors, or just wealthy?

Statistical Detection: The Z-Score Method

The Formula

Measures how many standard deviations a data point is from the mean.

$$Z = \frac{X - \mu}{\sigma}$$

Threshold: Typically, $|Z| > 3$ is considered an outlier.

Results for Account Balance

- Outliers flagged: 177

The Discrepancy

Z-score flagged significantly fewer outliers (177) compared to IQR (503). Why?

The Normality Assumption

The Z-score method assumes the data is approximately **normally distributed** (bell-shaped).

Why it fails on skewed data

In highly right-skewed financial data, the mean (μ) is pulled toward the tail, and the standard deviation (σ) is massively inflated by those same extreme values. This artificially widens the threshold, allowing many extreme values to “hide” within 3 standard deviations.

Takeaway

For skewed financial variables, IQR is mathematically superior to Z-score for outlier detection.

Part 3: Outlier Treatment Decisions

"Do not delete what you do not understand."

The Golden Rule: Outlier $\neq$ Error

The Common Mistake

Beginner analysts often apply a blanket rule: “Delete everything beyond the 95th percentile.” This is statistically dangerous and business-blind.

The Financial Reality

- A SACCO member with a 5,000,000 KES balance is likely a **high-net-worth individual**, an SME, or an institutional investor.
- Deleting them removes your most valuable customers from the dataset.
- This introduces **selection bias**, causing the model to underrepresent critical business segments.

[The Core Distinction] An outlier is a statistical concept (a value far from the center). An error is a data-quality concept (a value that is impossible). **Treat them differently.**

The Trainer's Question

Before removing an extreme observation, always ask:
“Is this value impossible, or is it simply unusual?”

The Outlier Decision Framework

Use this domain-guided framework to decide the fate of every flagged outlier:

Category	Example	Action
Data-entry error	Age = 150, Balance = -500	Correct if possible, otherwise remove .
Impossible value	Tenure > Customer Lifetime	Investigate logic, correct or remove .
Valid but extreme	Balance = 2,000,000 KES	Retain , but apply capping or transformation.
Business-critical	Institutional SME account	Retain and carefully assess model influence.

Key Principle

Outlier treatment must be guided by **statistical evidence and domain knowledge**, never by an arbitrary statistical threshold alone.

Treatment Strategy: Capping (Winsorization)

The Concept

Instead of deleting valid extreme observations, we **limit their influence** by replacing values above a certain threshold with the threshold value itself.

The Mechanics

- Calculate the 95th percentile of the variable (e.g., 500,000 KES).
- Any customer with a balance $> 500,000$ KES is assigned exactly 500,000 KES for the purpose of modelling.
- The original value is not deemed “wrong”; it is simply compressed.

Why the 95th Percentile?

The 95th percentile is a practical, business-friendly choice. It compresses the top 5% of extremes while leaving the bottom 95% of the distribution completely untouched. (Note: The threshold should always be justified by the specific business context).

Crucial Clarification

Capping does *not* mean the original value was an error. It is a modelling strategy used to prevent extreme values from dominating distance calculations and gradient updates.

Advantages of Capping over Deletion

1. Preserves Sample Size

No customers are removed from the dataset. You retain all 10,000 records, maintaining maximum statistical power.

2. Reduces Disproportionate Influence

Distance-based algorithms (KNN, K-Means, SVM) calculate geometric distances. A balance of 5,000,000 would mathematically overpower an age of 40. Capping neutralizes this geometric distortion.

3. Maintains Segment Representation

High-net-worth individuals remain in the dataset, allowing the model to learn the churn behaviors of the SME/institutional segment, rather than ignoring them entirely.

Visual Confirmation

After capping the Balance and Age variables at their respective 95th percentiles, regenerated boxplots show the “dots” (outliers) disappearing. The upper whisker now cleanly represents the compressed boundary of the data.

The Result

The data is now “model-safe” for distance-sensitive algorithms, without sacrificing business reality.

Part 4: Feature Encoding Strategies

"Translating business categories into mathematical signals."

The Categorical Challenge

The Problem

Machine learning algorithms require numerical matrices. They cannot natively process text strings like “Male”, “Nairobi”, or “University”.

The Risk

If we simply map text to arbitrary numbers (e.g., Nairobi=1, Mombasa=2, Kisumu=3), the model will assume that Kisumu is “three times greater” than Nairobi, and that Mombasa is the “average” of Nairobi and Kisumu. This injects **false mathematical relationships**.

The Solution

We must choose an encoding strategy that respects the **cardinality** (number of unique categories) and the **hierarchy** (natural order) of the variable.

Three Main Strategies

- 1 **Ordinal Encoding:** For variables with a natural hierarchy.
- 2 **One-Hot Encoding:** For nominal variables with low cardinality.
- 3 **Target Encoding:** For nominal variables with high cardinality.

The Golden Rule of Encoding

Never let the encoding algorithm invent a mathematical relationship that doesn't exist in the real world.

Ordinal Encoding: Preserving Hierarchy

When to use it

When categories have a meaningful, natural order or progression.

Example: Education Level

- None $\rightarrow$ 0
- Primary $\rightarrow$ 1
- Secondary $\rightarrow$ 2
- University $\rightarrow$ 3

Why it works

The model correctly interprets that a University degree represents a higher level of educational attainment than Primary school. The numerical distance carries semantic meaning.

Advantages

- Computationally efficient (creates only 1 column).
- Preserves the logical hierarchy of the data.
- Works well with tree-based models and linear models.

Warning

Do *not* use Ordinal Encoding for nominal variables like “Color” (Red, Blue, Green) or “County”. Assigning 1, 2, 3 to colors implies $\text{Green} > \text{Blue} > \text{Red}$, which is mathematically nonsensical.

One-Hot Encoding: Low Cardinality

When to use it

For nominal variables (no natural order) that have a **small number of unique categories** (low cardinality).

Example: Gender & Marital Status

- Gender_Male: 0 or 1
- MaritalStatus_Married: 0 or 1
- MaritalStatus_Single: 0 or 1

The Mechanics

Creates a new binary (0/1) dummy column for each category. We drop the first category (`drop_first=True`) to avoid the “dummy variable trap” (perfect multicollinearity).

Advantages

- Does not impose false ordinal relationships.
- Easily interpreted by linear models and neural networks.

The Cardinality Limit

If applied to a variable with 47 Counties, One-Hot Encoding creates 46 new columns. This leads to sparse matrices, the curse of dimensionality, and severe overfitting.

Target Encoding: The High Cardinality Solution

When to use it

For nominal variables with **many unique categories** (e.g., 47 Counties, 100 Branch IDs).

The Concept

Replace the text category with a numerical value based on its historical relationship with the **target variable**.

Example: County Churn Rate

County	Encoded Value (Historical Churn)
Nairobi	0.32
Mombasa	0.27
Kisumu	0.21

Why it works

Instead of creating 46 sparse dummy columns, we create **one dense, highly informative column**. The model immediately knows that a customer from Nairobi has a historically higher risk of churning than one from Kisumu.

The Massive Risk

If not implemented carefully, Target Encoding causes severe **data leakage**, allowing the model to “cheat” by looking at the answer key during training.

The Danger of Target Leakage

What is Target Leakage?

Leakage occurs when the model uses information during training that it would not have access to in a real-world prediction scenario.

The Naive Mistake

If you calculate the mean churn rate for Nairobi using the *entire* dataset, you are including the target value of the specific customer you are currently trying to predict.

The Consequence

The model memorizes the exact churn rates of the training data. It achieves 99% accuracy in training, but fails catastrophically on unseen test data because it was “fed the answers.”

The Golden Rule of Target Encoding

Never use the target value of the current row to encode the current row.

The Solution

To prevent leakage, we must calculate the target statistic using *out-of-sample* data. This is achieved through **K-Fold Cross-Validation Target Encoding**.

K-Fold Target Encoding: Preventing Leakage

The Mechanics

- 1 Split the training data into K folds (e.g., $K = 5$).
- 2 For Fold 1, calculate the mean churn rate of each county using **only Folds 2, 3, 4, and 5**.
- 3 Apply these calculated means to encode the counties in Fold 1.
- 4 Repeat this process for all K folds.

The Result

Every row is encoded using data from *other* rows. No row's target value is used to encode itself.

Why it is robust

- Strictly prevents target leakage.
- Acts as a form of regularization (smoothing), preventing the model from overfitting to categories with very few samples.
- Reduces dimensionality while retaining high predictive signal.

Trainer Emphasis

High cardinality → Target Encoding reduces dimensions
→ K-Fold implementation prevents leakage. This is the professional standard for encoding geographical or branch-level data.

Part 5: Scaling, Normalization & Statistical Thinking

"Does the algorithm actually care about the scale?"

Why Scale Features?

The Problem of Magnitude

Financial balances range from 0 to 1,000,000+. Customer ages range from 18 to 80.

Distance-Sensitive Models

Algorithms like **KNN**, **SVM**, **K-Means**, and **Neural Networks** rely on geometric distance (e.g., Euclidean distance).

- A difference of 10,000 KES in balance will mathematically overpower a difference of 10 years in age.
- The model will effectively ignore the age variable.

Gradient-Descent Models

Logistic Regression and Neural Networks converge much faster and more stably when features are on similar scales.

Tree-Based Models (The Exception)

Decision Trees, **Random Forests**, and **XGBoost** do *not* require scaling.

Why Trees Don't Care

Trees make decisions using orthogonal thresholds:
"Is Balance > 100,000?"

They do not calculate distances between points. Scaling a feature from $[0, 1M]$ to $[0, 1]$ changes nothing about the tree's split logic. Scaling for tree models wastes computation.

Comparing Scalers: Standard vs MinMax vs Robust

Scaler	Mechanism	Sensitivity to Outliers	Best For
StandardScaler	Centers to Mean=0, scales to Std=1	High. Mean and Std are easily pulled by extremes.	Normally distributed data.
MinMaxScaler	Scales data to a fixed range, usually [0, 1]	Very High. Outliers compress the rest of the data into a tiny range.	Image processing, bounded data.
RobustScaler	Centers to Median, scales using IQR	Low. Median and IQR ignore extreme tails.	Skewed financial data, data with outliers.

The MinMax Trap

If a single customer has a balance of 50M KES, MinMaxScaler will squash the other 9,999 customers into a range of 0.000 to 0.020, destroying their variance.

The Robust Solution

RobustScaler uses the exact same logic as our outlier detection (Median and IQR). It is mathematically aligned with skewed financial distributions.

The RobustScaler Advantage for Financial Data

How it works

$$X_{scaled} = \frac{X - \text{Median}}{\text{IQR}}$$

Results on Capped Balance

- **StandardScaler:** Mean = 0.00, Std = 1.00
- **MinMaxScaler:** Mean = 0.31, Std = 0.28
- **RobustScaler:** Mean = 0.24, Std = 0.77

Why it wins here

Even after capping at the 95th percentile, financial data retains skewness. RobustScaler ensures the bulk of the customers (the middle 50%) are spread out meaningfully, without being distorted by the capped extremes.

Statistical Thinking

Scaling is not a “one-size-fits-all” preprocessing step that you blindly apply to every column. It is a mathematical intervention required *only* when the algorithm’s distance metric demands it.

The Trainer’s Question

Before scaling, always ask:

“Does this specific algorithm care about the scale of the predictors?”

If the answer is no (e.g., XGBoost), skip the scaler.

Statistical Clarification: The Normality Myth

The Common Misconception

Many analysts believe they must apply `StandardScaler` to force data into a normal (Gaussian) bell curve, or that Logistic Regression requires normally distributed predictors.

The Statistical Reality

- **StandardScaler does NOT make data normal.** It only shifts the mean to 0 and scales the variance. A skewed distribution remains skewed after StandardScaling.
- **Logistic Regression does NOT assume normal predictors.** It assumes a linear relationship between the log-odds of the target and the predictors.

Why Do We Scale Then?

We scale for **numerical stability**, comparability of coefficients (in linear models), and optimization speed (in gradient descent), *not* to satisfy a normality assumption.

How to Actually Fix Skewness

If an algorithm genuinely requires normally distributed features (e.g., Linear Discriminant Analysis), you must use mathematical transformations like the **Log Transform** or **Box-Cox Transform**, not a scaler.

Building the Final ML Pipeline

Selecting the Clean Features

After preprocessing, we select the final model-ready columns.

Our Final Selection

- **Numerical:** Age, Tenure, NumProducts, SaccoScore, MonthsInactive, BranchVisits, UtilizationRate
- **Encoded:** Education_Ord, EmploymentStatus_TE
- **Scaled:** Balance_RobustScaler
- **Dummies:** Gender_Male, MaritalStatus dummies
- **Target:** Churn

The Preprocessing Justification

Every technique was selected based on data characteristics:

- **Capping:** Retained SMEs while neutralizing distance dominance.
- **Ordinal/Target Encoding:** Handled hierarchies and high-cardinality without dimensionality explosion.
- **RobustScaler:** Aligned with the skewed nature of financial balances.

Deliverable

The transformed dataset is exported as `churn_transformed_session2.csv`, fully prepped for Week 2 modelling.

Key Takeaways

- **Outliers $\neq$ Errors:** Use domain knowledge to distinguish between data-entry mistakes and valid high-net-worth extremes.
- **Cap, Don't Delete:** Winsorization (capping) preserves sample size and segment representation while neutralizing geometric dominance in distance-based models.
- **Match Encoding to Cardinality:** Use Ordinal for hierarchies, One-Hot for low-cardinality nominals, and K-Fold Target Encoding for high-cardinality variables to prevent leakage.
- **Beware Target Leakage:** Never use a row's own target value to encode its features. K-Fold encoding is the professional safeguard.
- **Scale Algorithmically:** Only scale when the model demands it (KNN, SVM, Neural Nets). Use **RobustScaler** for skewed financial data; skip scaling entirely for Tree-based models.
- **Dispelling the Myth:** Scalers do not make data normal, and Logistic Regression does not require normal predictors. Scale for numerical stability, not to satisfy a false assumption.

Best Practices Checklist

Stage	Action	Rationale
Outlier Detection	Use IQR for skewed data, Z-score for normal data.	IQR is robust to the long tails common in finance.
Outlier Treatment	Cap at 95th/99th percentile.	Retains valuable customers; prevents distance distortion.
Low-Cardinality	One-Hot Encoding (drop first).	Prevents false ordinal relationships and multicollinearity.
High-Cardinality	K-Fold Target Encoding.	Reduces dimensions while strictly preventing target leakage.
Scaling	RobustScaler for skewed features.	Uses Median/IQR, ignoring extreme capped values.
Tree Models	Skip scaling entirely.	Trees use thresholds, not distances; scaling is wasted compute.

The Ultimate Mindset

Understand the data → Identify the problem → Select the transformation → Justify the decision → Validate the result.

Session Summary

What We Covered Today

- Loaded and profiled a 10,000-row Kenyan SACCO churn dataset.
- Detected outliers using Boxplots, IQR, and Z-score methods.
- Applied Winsorization (capping) to preserve high-net-worth customers.
- Encoded variables using Ordinal, One-Hot, and K-Fold Target strategies.
- Compared Standard, MinMax, and Robust scalers, aligning them with algorithm sensitivity.
- Debunked the “normality assumption” myth in scaling.

Coming Up in Week 2

- **Next session:** Feature Selection, Dimensionality Reduction (PCA), and handling multi-collinearity.
- **Later:** Building the actual predictive models (Logistic Regression, Random Forest, XGBoost) on this preprocessed data.

Preparation

Review the K-Fold Target Encoding logic. Think about how you would handle a dataset with 500 unique branch IDs. Bring questions.

Session Deliverable

The Preprocessing Memo

Your deliverable for this session is a technical memo documenting the preprocessing pipeline applied to the SACCO dataset.

Required Sections:

- 1 **Outlier Strategy:** Justify why capping was chosen over deletion for the Balance variable.
- 2 **Encoding Strategy:** Explain why K-Fold Target Encoding was used for EmploymentStatus/County, and how it prevents leakage.
- 3 **Scaling Strategy:** Justify the choice of RobustScaler over StandardScaler for financial features.
- 4 **Algorithm Alignment:** State which models require this scaling and which do not.

The Standard of Excellence

A professional data scientist does not just write code; they write **defensible documentation**. Every preprocessing choice must be justified by statistical theory and business context.

Output

The final, model-ready dataset (`churn_transformed_session2.csv`) must be submitted alongside the memo, proving that the pipeline executed successfully without data leakage.

Questions?

*Let's discuss IQR vs Z-score, when to cap vs delete,
how K-Fold Target Encoding prevents leakage, or algorithm scaling rules.*

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